

THE UNIVERSITY OF WESTERN AUSTRALIA

Computer Science & Software Engineering

SEMESTER 1, 2016 EXAMINATIONS

CITS4403
Computational Modelling

FAMILY NAME: _____ GIVEN NAMES: _____

STUDENT ID: | | | | | | | | | SIGNATURE: _____

This Paper Contains: **15** pages **(including title page)**
Time allowed: **2:00 hours**

INSTRUCTIONS:

There are 22 questions. Marks for this paper total 120, Candidate must answer all questions.

Answers are to be written on this question paper in the spaces provided.

Calculators are not permitted.

PLEASE NOTE

Examination candidates may only bring authorised materials into the examination room. If a supervisor finds, during the examination, that you have unauthorised material, in whatever form, in the vicinity of your desk or on your person, whether in the examination room or the toilets or en route to/from the toilets, the matter will be reported to the head of school and disciplinary action will normally be taken against you. This action may result in your being deprived of any credit for this examination or even, in some cases, for the whole unit. This will apply regardless of whether the material has been used at the time it is found.

Therefore, any candidate who has brought any unauthorised material whatsoever into the examination room should declare it to the supervisor immediately. Candidates who are uncertain whether any material is authorised should ask the supervisor for clarification.

Supervisors Only - Student left at:

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(Questions 1 to 20 carry 5 marks each)

1. Explain how software libraries and their dependencies could be represented as a graph. What properties would you expect this graph to have.

[5 marks]

2. Give two different methods for generating random graphs, and explain the difference in the types of graphs that they generate.

[5 marks]

3. Describe an algorithm for determining the average shortest path length between any two vertices in a graph.

[5 marks]

4. What is the order of growth of the algorithm you described in question 3.

[5 marks]

5. Describe an efficient implementation of the Dictionary (or Map) data structure. What is the complexity of the three main operations.

[5 marks]

6. Describe Schelling's model of racial segregation, and what its significance is.

[5 marks]

7. Describe the long tail property and give two different systems where this property may arise.

[5 marks]

8. What is the demarcation problem. Explain how Popper proposed to address the demarcation problem.

[5 marks]

9. Do you consider Zipf's law to be an explantory model, and why?

[5 marks]

10. What are the characteristic properties of a Pareto distribution?

[5 marks]

11. Explain the differences between randomness and complexity?

[5 marks]

12. What is the Church-Turing thesis and how is it related to universality?

[5 marks]

13. Describe the four main classes of one dimensional cellular automata.

[5 marks]

14. "In the game of life, all initial states eventually reach a simple repeating pattern." Discuss whether this claim is true or false.

[5 marks]

15. How are scientific realism and instrumentalism related to reductionism and holism?

[5 marks]

16. Define fractal dimension and give and example of something with non-integer dimension.

[5 marks]

17. What does it mean to say a system is computationally irreducible?

[5 marks]

18. What is the fast Fourier transform, and what is its significance?

[5 marks]

19. Describe the sand pile model of Bak, Tang and Wiesenfield, and what can we infer from it.

[5 marks]

20. Give three characteristics of agent based models.

[5 marks]

21. Describe how a two dimensional cellular automaton could be used to model crowd dynamics in a crowded train station.

[10 marks]

22. Do you think the complexity of systems can be objectively measured? Present a reasoned argument for your answer.

[10 marks]

END OF PAPER

Blank page for extra writing if needed